

OSSP (Operating Systems And Systems Programming) - 25CS2104E

(SKILL Content Delivery Concepts)

1. To Install Linux VM, Configure GCC, Setup Git Repository, Create Project Structure, Understand Shell Architecture, Build Initial Makefile

To Analyze process abstraction, execute `fork()`, understand `exec()` family, analyze parent-child relationships, inspect process tree, practice system call tracing

```
mokshitha_putti@MokshithaF × + ~$ sudo apt update
[sudo: authenticate] Password:
Hit:1 http://archive.ubuntu.com/ubuntu resolute InRelease
Get:2 http://security.ubuntu.com/ubuntu resolute-security InRelease [137
Get:3 http://archive.ubuntu.com/ubuntu resolute-updates InRelease [137
Get:4 http://security.ubuntu.com/ubuntu resolute-security/main amd64 P
s [478 kB]
Get:5 http://archive.ubuntu.com/ubuntu resolute-backports InRelease [137
Get:6 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Pac
[612 kB]
Get:7 http://security.ubuntu.com/ubuntu resolute-security/main Transla
n [113 kB]
Get:8 http://security.ubuntu.com/ubuntu resolute-security/main amd64 C
nts [46.6 kB]
Get:9 http://security.ubuntu.com/ubuntu resolute-security/universe am
```

```
mokshitha_putti@MokshithaPutti:~$ sudo apt install build-essential git strace tree -y
build-essential is already the newest version (12.12ubuntu2.26.04.2).
git is already the newest version (1:2.53.0-1ubuntu1).
strace is already the newest version (6.19+ds-0ubuntu5).
Installing:
  tree

Summary:
  Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 91
  Download size: 53.5 kB
  Space needed: 125 kB / 1024 GB available

Get:1 http://archive.ubuntu.com/ubuntu resolute/universe amd64 tree amd64 2.3.1-1 [53.5 kB]
Fetched 53.5 kB in 2s (21.6 kB/s)
Selecting previously unselected package tree.
(Reading database ... 43117 files and directories currently installed.)
Preparing to unpack .../tree_2.3.1-1_amd64.deb ...
Unpacking tree (2.3.1-1) ...
Setting up tree (2.3.1-1) ...
Processing triggers for man-db (2.13.1-1build1) ...
mokshitha_putti@MokshithaPutti:~$
```

```
#include <stdio.h>

int main(void)
{
    printf("OSSP Mini Shell - Project Started\n");
    printf("Question 1 setup completed.\n");

    return 0;
}
```

```

mokshitha_putti@MokshithaPutti:~/ossp-mini-shell$ git init
hint: Using 'master' as the name for the initial branch. This default branch
name
hint: will change to "main" in Git 3.0. To configure the initial branch name
hint: to use in all of your new repositories, which will suppress this warni
ng,
hint: call:
hint:     git config --global init.defaultBranch <name>
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command
:
hint:     git branch -m <name>
hint: Disable this message with "git config set advice.defaultBranchName fal
se"
Initialized empty Git repository in /home/mokshitha_putti/ossp-mini-shell/.g
it/
mokshitha_putti@MokshithaPutti:~/ossp-mini-shell$ |

```

```

mokshitha_putti@MokshithaPutti:~/ossp-mini-shell$ |
hint: call:
hint:     git config --global init.defaultBranch <name>
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command
:
hint:     git branch -m <name>
hint: Disable this message with "git config set advice.defaultBranchName fal
se"
Initialized empty Git repository in /home/mokshitha_putti/ossp-mini-shell/.g
it/
mokshitha_putti@MokshithaPutti:~/ossp-mini-shell$ nano src/main.c
mokshitha_putti@MokshithaPutti:~/ossp-mini-shell$ gcc src/main.c -o ossp_she
ll
mokshitha_putti@MokshithaPutti:~/ossp-mini-shell$ ./ossp_shell
OSSP Mini Shell - Project Started
Question 1 setup completed.
mokshitha_putti@MokshithaPutti:~/ossp-mini-shell$ nano Makefile
mokshitha_putti@MokshithaPutti:~/ossp-mini-shell$ make clean
rm -f ossp_shell
mokshitha_putti@MokshithaPutti:~/ossp-mini-shell$ make
gcc -Wall -Wextra -std=c11 src/main.c -o ossp_shell
mokshitha_putti@MokshithaPutti:~/ossp-mini-shell$ make run
./ossp_shell
OSSP Mini Shell - Project Started
Question 1 setup completed.
mokshitha_putti@MokshithaPutti:~/ossp-mini-shell$ tree
.
├── Makefile
├── docs
├── include
├── ossp_shell
├── results
├── src
│   └── main.c
└── tests

6 directories, 3 files

```

```
mokshitha_putti@MokshithaPutti:~$ sudo apt install build-essential git strace
tree -y
build-essential is already the newest version (12.12ubuntu2.26.04.2).
git is already the newest version (1:2.53.0-1ubuntu1).
strace is already the newest version (6.19+ds-0ubuntu5).
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Preparing to unpack .../tree_2.3.1-1_amd64.deb ...
Unpacking tree (2.3.1-1) ...
Setting up tree (2.3.1-1) ...
Processing triggers for man-db (2.13.1-1build1) ...
mokshitha_putti@MokshithaPutti:~$ mkdir -p ~/ossp-mini-shell
cd ~/ossp-mini-shell
mkdir -p src include tests docs results
mokshitha_putti@MokshithaPutti:~/ossp-mini-shell$ tree
.
├── docs
├── include
├── results
├── src
└── tests

6 directories, 0 files
mokshitha_putti@MokshithaPutti:~/ossp-mini-shell$ git init
hint: Using 'master' as the name for the initial branch. This default branch
name
hint: will change to "main" in Git 3.0. To configure the initial branch name
hint: to use in all of your new repositories, which will suppress this warni
ng,
hint: call:
```

```
GNU nano 8.7.1 src/process_demo.c *
exit(1);
}
else
{
    /* Parent process */
    printf("\n--- PARENT PROCESS ---\n");
    printf("Parent PID: %d\n", getpid());
    printf("Child PID: %d\n", pid);

    printf("Parent is waiting for child...\n");

    waitpid(pid, NULL, 0);

    printf("Child finished.\n");
    printf("Parent continues execution.\n");
}

return 0;
}|
```

```

mokshitha_putti@Mokshithaf x + v
-rwxr-xr-x 1 mokshitha_putti mokshitha_putti 16304 Sep 17 15:44 process_demo
drwxr-xr-x 2 mokshitha_putti mokshitha_putti 4096 Sep 17 15:28 results
drwxr-xr-x 2 mokshitha_putti mokshitha_putti 4096 Sep 17 15:44 src
drwxr-xr-x 2 mokshitha_putti mokshitha_putti 4096 Sep 17 15:28 tests
Child finished.
Parent continues execution.
mokshitha_putti@MokshithaPutti:~/oss-mini-shell$ ./process_demo
=== PROCESS DEMONSTRATION ===
Parent PID: 1081

--- PARENT PROCESS ---
Parent PID: 1081
Child PID: 1082
Parent is waiting for child...

--- CHILD PROCESS ---
Child PID: 1082
Parent PID: 1081
Child is executing the 'ls' program...
total 56
-rw-r--r-- 1 mokshitha_putti mokshitha_putti 196 Sep 17 15:40 Makefile
drwxr-xr-x 2 mokshitha_putti mokshitha_putti 4096 Sep 17 15:28 docs
drwxr-xr-x 2 mokshitha_putti mokshitha_putti 4096 Sep 17 15:28 include
-rwxr-xr-x 1 mokshitha_putti mokshitha_putti 15952 Sep 17 15:40 ossp_shell
-rwxr-xr-x 1 mokshitha_putti mokshitha_putti 16304 Sep 17 15:44 process_demo
drwxr-xr-x 2 mokshitha_putti mokshitha_putti 4096 Sep 17 15:28 results
drwxr-xr-x 2 mokshitha_putti mokshitha_putti 4096 Sep 17 15:44 src
drwxr-xr-x 2 mokshitha_putti mokshitha_putti 4096 Sep 17 15:28 tests
Child finished.
Parent continues execution.
mokshitha_putti@MokshithaPutti:~/oss-mini-shell$ ls -l
total 56
-rw-r--r-- 1 mokshitha_putti mokshitha_putti 196 Sep 17 15:40 Makefile
drwxr-xr-x 2 mokshitha_putti mokshitha_putti 4096 Sep 17 15:28 docs
drwxr-xr-x 2 mokshitha_putti mokshitha_putti 4096 Sep 17 15:28 include
-rwxr-xr-x 1 mokshitha_putti mokshitha_putti 15952 Sep 17 15:40 ossp_shell
-rwxr-xr-x 1 mokshitha_putti mokshitha_putti 16304 Sep 17 15:44 process_demo
drwxr-xr-x 2 mokshitha_putti mokshitha_putti 4096 Sep 17 15:28 results
drwxr-xr-x 2 mokshitha_putti mokshitha_putti 4096 Sep 17 15:44 src
drwxr-xr-x 2 mokshitha_putti mokshitha_putti 4096 Sep 17 15:28 tests
mokshitha_putti@MokshithaPutti:~/oss-mini-shell$

```

```

--- PARENT PROCESS ---
--- CHILD PROCESS ---
Parent PID: 1155
Child PID: 1156
Child PID: 1156
Parent is waiting for child...
[pid 1155] wait4(1156, &status, 0, NULL) = 1156
Child is executing the 'ls' program...
<unfinished ...>
[pid 1156] execve("/home/mokshitha_putti/bin/ls", ["ls", "-l"], 0x7ffd18d546d8 /* 27 vars */) = -1 ENOENT (No such file or directory)
[pid 1156] execve("/usr/local/sbin/ls", ["ls", "-l"], 0x7ffd18d546d8 /* 27 vars */) = -1 ENOENT (No such file or directory)
[pid 1156] execve("/usr/local/bin/ls", ["ls", "-l"], 0x7ffd18d546d8 /* 27 vars */) = -1 ENOENT (No such file or directory)
[pid 1156] execve("/usr/sbin/ls", ["ls", "-l"], 0x7ffd18d546d8 /* 27 vars */) = -1 ENOENT (No such file or directory)
[pid 1156] execve("/usr/bin/ls", ["ls", "-l"], 0x7ffd18d546d8 /* 27 vars */) = 0
total 56
-rw-r--r-- 1 mokshitha_putti mokshitha_putti 196 Sep 17 15:40 Makefile
drwxr-xr-x 2 mokshitha_putti mokshitha_putti 4096 Sep 17 15:28 docs
drwxr-xr-x 2 mokshitha_putti mokshitha_putti 4096 Sep 17 15:28 include
-rwxr-xr-x 1 mokshitha_putti mokshitha_putti 15952 Sep 17 15:40 ossp_shell
-rwxr-xr-x 1 mokshitha_putti mokshitha_putti 16304 Sep 17 15:44 process_demo
drwxr-xr-x 2 mokshitha_putti mokshitha_putti 4096 Sep 17 15:28 results
drwxr-xr-x 2 mokshitha_putti mokshitha_putti 4096 Sep 17 15:44 src
drwxr-xr-x 2 mokshitha_putti mokshitha_putti 4096 Sep 17 15:28 tests
[pid 1156] exit_group(0) = ?
[pid 1156] +++ exited with 0 +++
<... wait4 resumed>, NULL, 0, NULL) = 1156
--- SIGCHLD {si_signo=SIGCHLD, si_code=CLD_EXITED, si_pid=1156, si_uid=1000, si_status=0, si_utime=0, si_stime=4 /* 0.04 s */} ---
Child finished.
Parent continues execution.
exit_group(0) = ?

```

```

mokshitha_putti@MokshithaPutti:~/ossp-mini-shell$ strace -f -e trace=process
./process_demo
execve("./process_demo", ["/process_demo"], 0x7ffffdd9b0388 /* 27 vars */) =
0
=== PROCESS DEMONSTRATION ===
Parent PID: 1155
clone(child_stack=NULL, flags=CLONE_CHILD_CLEARTID|CLONE_CHILD_SETTID|SIGCHL
Dstrace: Process 1156 attached
, child_tidptr=0x786cc5307a10) = 1156
--- PARENT PROCESS ---
--- CHILD PROCESS ---
Parent PID: 1155
Child PID: 1156
Child PID: 1156
Parent is waiting for child...
[pid 1155] wait4(1156Parent PID: 1155
Child is executing the 'ls' program...
<unfinished ...>
[pid 1156] execve("/home/mokshitha_putti/bin/ls", ["ls", "-l"], 0x7fffd18d54
6d8 /* 27 vars */) = -1 ENOENT (No such file or directory)
[pid 1156] execve("/usr/local/sbin/ls", ["ls", "-l"], 0x7fffd18d546d8 /* 27
vars */) = -1 ENOENT (No such file or directory)
[pid 1156] execve("/usr/local/bin/ls", ["ls", "-l"], 0x7fffd18d546d8 /* 27 v
ars */) = -1 ENOENT (No such file or directory)
[pid 1156] execve("/usr/sbin/ls", ["ls", "-l"], 0x7fffd18d546d8 /* 27 vars *
/) = -1 ENOENT (No such file or directory)
[pid 1156] execve("/usr/bin/ls", ["ls", "-l"], 0x7fffd18d546d8 /* 27 vars */
) = 0
total 56
-rw-r--r-- 1 mokshitha_putti mokshitha_putti 196 Sep 17 15:40 Makefile
drwxr-xr-x 2 mokshitha_putti mokshitha_putti 4096 Sep 17 15:28 docs
drwxr-xr-x 2 mokshitha_putti mokshitha_putti 4096 Sep 17 15:28 include
-rwxr-xr-x 1 mokshitha_putti mokshitha_putti 15952 Sep 17 15:40 ossp_shell
-rwxr-xr-x 1 mokshitha_putti mokshitha_putti 16304 Sep 17 15:44 process_demo
drwxr-xr-x 2 mokshitha_putti mokshitha_putti 4096 Sep 17 15:28 results
drwxr-xr-x 2 mokshitha_putti mokshitha_putti 4096 Sep 17 15:44 src
drwxr-xr-x 2 mokshitha_putti mokshitha_putti 4096 Sep 17 15:28 tests
[pid 1156] exit_group(0) = ?
[pid 1156] +++ exited with 0 +++

```

```

drwxr-xr-x 2 mokshitha_putti mokshitha_putti 4096 Sep 17 15:28 tests
[pid 1156] exit_group(0) = ?
[pid 1156] +++ exited with 0 +++
<... wait4 resumed>, NULL, 0, NULL) = 1156
--- SIGCHLD {si_signo=SIGCHLD, si_code=CLD_EXITED, si_pid=1156, si_uid=1000,
si_status=0, si_utime=0, si_stime=4 /* 0.04 s */} ---
Child finished.
Parent continues execution.
exit_group(0) = ?
+++ exited with 0 +++
mokshitha_putti@MokshithaPutti:~/ossp-mini-shell$ git add .
mokshitha_putti@MokshithaPutti:~/ossp-mini-shell$ git commit -m "Complete Qu
estion 1 process fundamentals"
Author identity unknown

*** Please tell me who you are.

Run

    git config --global user.email "you@example.com"
    git config --global user.name "Your Name"

to set your account's default identity.
Omit --global to set the identity only in this repository.

fatal: empty ident name (for <mokshitha_putti@MokshithaPutti.localdomain>) n
ot allowed
mokshitha_putti@MokshithaPutti:~/ossp-mini-shell$ git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
    new file:   Makefile
    new file:   ossp_shell
    new file:   process_demo
    new file:   src/main.c
    new file:   src/process_demo.c
mokshitha_putti@MokshithaPutti:~/ossp-mini-shell$

```

2. To Create Main Loop, Display Prompt, Read User Input, Handle Exit Conditions, Design Control Flow Diagram, Test Interactive Loop

To Capture Keyboard Input, Handle Backspace, Process Enter Key, Manage Input Buffer, Support Multi-Character Commands, Test User Interaction

```
mokshitha_putti@MokshithaPutti:~$ cd ~/ossp-mini-shell
mkdir -p src
nano src/main_loop.c
mokshitha_putti@MokshithaPutti:~/ossp-mini-shell$ gcc src/main_loop.c -o myshell
mokshitha_putti@MokshithaPutti:~/ossp-mini-shell$ ./myshell
myshell> hello
You entered: hello
myshell> test
You entered: test
myshell> exit
mokshitha_putti@MokshithaPutti:~/ossp-mini-shell$
```

3. To Apply Escape Sequences, Store Command History, Navigate Previous Commands, Navigate Next Commands, Update Input Buffer, Test Recall Functionality.

To Allocate Buffers Dynamically, Resize Arrays, Prevent Buffer Overflow, Manage Linked Lists, Release Memory Correctly, Verify with Valgrind.

```
mokshitha_putti@MokshithaPutti:~/ossp-mini-shell$ cd ~/ossp-mini-shell
nano src/history.c
mokshitha_putti@MokshithaPutti:~/ossp-mini-shell$ gcc src/history.c -o history
./history
myshell> hello
You entered: hello
myshell> linux
You entered: linux
myshell> linux
You entered: linux
myshell> linux
You entered: linux
myshell> hello
You entered: hello
myshell> exit
mokshitha_putti@MokshithaPutti:~/ossp-mini-shell$ |
```

```

myshell> exit
mokshitha_putti@MokshithaPutti:~/ossp-mini-shell$ sudo apt update
sudo apt install valgrind -y
[sudo: authenticate] Password:
Hit:1 http://archive.ubuntu.com/ubuntu resolute InRelease
Get:2 http://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB]
Get:3 http://archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Hit:4 http://archive.ubuntu.com/ubuntu resolute-backports InRelease
Get:5 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Packages [478 kB]
Get:6 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Packages [615 kB]
Get:7 http://archive.ubuntu.com/ubuntu resolute-updates/main Translation-en [148 kB]
Get:8 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Packages [294 kB]
Get:9 http://archive.ubuntu.com/ubuntu resolute-updates/restricted amd64 Packages [413 kB]
Get:10 http://archive.ubuntu.com/ubuntu resolute-updates/restricted Translation-en [80.0 kB]
Fetched 2303 kB in 3s (875 kB/s)
92 packages can be upgraded. Run 'apt list --upgradable' to see them.
valgrind is already the newest version (1:3.26.0-0ubuntu1).
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 92
mokshitha_putti@MokshithaPutti:~/ossp-mini-shell$ valgrind --leak-check=full ./history
==709== Memcheck, a memory error detector
==709== Copyright (C) 2002-2024, and GNU GPL'd, by Julian Seward et al.
==709== Using Valgrind-3.26.0 and LibVEX; rerun with -h for copyright info
==709== Command: ./history
==709==
myshell> |

```

4. To Split Input into Tokens, Identify Delimiters, Handle Whitespace, Create Token Structures, Validate Token Streams, Debug Parsing Output.

To Design Parser Logic, Generate Parse Trees, Validate Syntax, Detect Errors, Handle Empty Commands, Produce Execution Structures.

```

mokshitha_putti@MokshithaPutti:~/ossp-mini-shell$ cd ~/ossp-mini-shell
nano src/parser.c
mokshitha_putti@MokshithaPutti:~/ossp-mini-shell$ gcc src/parser.c -o parser
mokshitha_putti@MokshithaPutti:~/ossp-mini-shell$ ./parser
myshell> ls -l /home
Total Tokens: 3
Token[0] = ls
Token[1] = -l
Token[2] = /home

myshell> pwd
Total Tokens: 1
Token[0] = pwd

myshell>
Error: Empty command

myshell> exit
mokshitha_putti@MokshithaPutti:~/ossp-mini-shell$

```

5. To Apply Single Quotes, Preserve Literal Content, Ignore Variable Expansion, Store Quoted Strings, Validate Parsing Results, Test Edge Cases.

To Apply Double Quotes, Preserve Spaces, Allow Variable Expansion, Parse Nested Tokens, Validate Outputs, Test Quoted Commands.

```

mokshitha_putti@MokshithaPutti:~/ossp-mini-shell$ cd ~/ossp-mini-shell
nano src/quotes.c
mokshitha_putti@MokshithaPutti:~/ossp-mini-shell$ gcc src/quotes.c -o quotes
mokshitha_putti@MokshithaPutti:~/ossp-mini-shell$ ./quotes
myshell> echo 'hello world'
Normal Token : echo
Single Quote : hello world

myshell> echo "hello world"
Normal Token : echo
Double Quote : hello world

myshell> echo "$USER"
Normal Token : echo
Double Quote : mokshitha_putti

myshell> echo 'ABC' "XYZ" "$HOME"
Normal Token : echo
Single Quote : ABC
Double Quote : XYZ
Double Quote : /home/mokshitha_putti

myshell>
Error: Empty command

myshell> |

```

6. To Create Process Escape Sequences, Handle Escaped Spaces, Escape Special Symbols, Preserve Characters, Validate Parser Output, Test Complex Inputs.

To Create Child Processes, Execute Programs, Handle Execution Errors, Pass Arguments, Manage Parent Process, Test Command Launching.

```

mokshitha_putti@MokshithaPutti:~/ossp-mini-shell$ cd ~/ossp-mini-shell
nano src/execute.c
mokshitha_putti@MokshithaPutti:~/ossp-mini-shell$ gcc src/execute.c -o execute
mokshitha_putti@MokshithaPutti:~/ossp-mini-shell$ ./execute
myshell> ls
Parsed Tokens:
[0] ls
Makefile      execute      include      ossp_shell   process_demo  results      tests
docs          history      myshell      parser       quotes
myshell> ls -l
Parsed Tokens:
[0] ls
[1] -l
total 144
-rw-r--r-- 1 mokshitha_putti mokshitha_putti 196 Sep 17 15:40 Makefile
drwxr-xr-x 2 mokshitha_putti mokshitha_putti 4096 Sep 17 15:28 docs
-rwxr-xr-x 1 mokshitha_putti mokshitha_putti 16664 Sep 17 18:57 execute
-rwxr-xr-x 1 mokshitha_putti mokshitha_putti 16632 Sep 17 18:39 history
drwxr-xr-x 2 mokshitha_putti mokshitha_putti 4096 Sep 17 15:28 include
-rwxr-xr-x 1 mokshitha_putti mokshitha_putti 16264 Sep 17 18:36 myshell
-rwxr-xr-x 1 mokshitha_putti mokshitha_putti 15952 Sep 17 15:40 ossp_shell
-rwxr-xr-x 1 mokshitha_putti mokshitha_putti 16344 Sep 17 18:50 parser
-rwxr-xr-x 1 mokshitha_putti mokshitha_putti 16304 Sep 17 15:44 process_demo
-rwxr-xr-x 1 mokshitha_putti mokshitha_putti 16344 Sep 17 18:53 quotes
drwxr-xr-x 2 mokshitha_putti mokshitha_putti 4096 Sep 17 15:28 results
drwxr-xr-x 2 mokshitha_putti mokshitha_putti 4096 Sep 17 18:57 src
drwxr-xr-x 2 mokshitha_putti mokshitha_putti 4096 Sep 17 15:28 tests
myshell> mkdir My\ Folder
Parsed Tokens:
[0] mkdir
[1] My Folder
myshell> touch file\#1.txt
Parsed Tokens:
[0] touch
[1] file\#1.txt
myshell> xyz123
Parsed Tokens:
[0] xyz123
Execution Error: No such file or directory
myshell> exit

```

7. Process Synchronization and Child Process Monitoring Using waitpid().

To Retrieve PATH Variable, Parse Search Directories, Locate Executables, Verify Permissions, Handle Missing Commands, Test Command Resolution.

```
mokshitha_putti@MokshithaPutti:~$ cd ~/ossp-mini-shell
nano src/waitpid_path.c
mokshitha_putti@MokshithaPutti:~/ossp-mini-shell$ gcc src/waitpid_path.c -o
waitpid_path
mokshitha_putti@MokshithaPutti:~/ossp-mini-shell$ ./waitpid_path
myshell> pwd
/home/mokshitha_putti/ossp-mini-shell
Child exited with status 0
myshell> showpath

PATH Directories:
/home/mokshitha_putti/bin
/usr/local/sbin
/usr/local/bin
/usr/sbin
/usr/bin
/sbin
/bin
/usr/games
/usr/local/games
/usr/lib/wsl/lib
/mnt/c/Program Files/WindowsApps/MicrosoftCorporationII.WindowsSubsystemForL
inux_2.6.1.0_x64__8wekyb3d8bbwe
/mnt/c/Program Files/Common Files/Oracle/Java/javapath
/mnt/c/Program Files (x86)/Common Files/Oracle/Java/java8path
/mnt/c/Program Files (x86)/Common Files/Oracle/Java/javapath
/mnt/c/Windows/system32
/mnt/c/Windows
/mnt/c/Windows/System32/Wbem
/mnt/c/Windows/System32/WindowsPowerShell/v1.0/
/mnt/c/Windows/System32/OpenSSH/
/mnt/c/Program Files/Docker/Docker/resources/bin
/mnt/c/Program Files/Git/cmd
/mnt/c/Program Files/nodejs/
/mnt/c/Program Files/MySQL/MySQL Shell 8.0/bin/
/mnt/c/Program Files/Common Files/Oracle/Java/javapath
/mnt/c/Program Files (x86)/Common Files/Oracle/Java/java8path
/mnt/c/Program Files (x86)/Common Files/Oracle/Java/javapath
/mnt/c/Windows/system32
/mnt/c/Windows
/mnt/c/Windows/System32/Wbem
/mnt/c/Program Files/Common Files/Oracle/Java/javapath
/mnt/c/Program Files (x86)/Common Files/Oracle/Java/java8path
/mnt/c/Program Files (x86)/Common Files/Oracle/Java/javapath
/mnt/c/Windows/system32
/mnt/c/Windows
/mnt/c/Windows/System32/Wbem
/mnt/c/Windows/System32/WindowsPowerShell/v1.0/
/mnt/c/Windows/System32/OpenSSH/
/mnt/c/Program Files/Docker/Docker/resources/bin
/mnt/c/Users/Mokshitha Putti/AppData/Local/Microsoft/WindowsApps
/mnt/c/Users/Mokshitha Putti/AppData/Local/Python/pythoncore-3.14-64/Script
/mnt/c/Users/Mokshitha Putti/AppData/Roaming/Python/Python314/Scripts
/mnt/c/users/mokshitha putti/.local/bin
/mnt/c/users/mokshitha putti/appdata/local/packages/pythonsoftwarefoundati
.python.3.12_qbz5n2kfra8p0/localcache/local-packages/python312/scripts
/mnt/c/Users/Mokshitha Putti/AppData/Local/Programs/Microsoft VS Code/bin
/mnt/c/Users/Mokshitha Putti/AppData/Roaming/npm
/snap/bin
*** stack smashing detected ***: terminated
Aborted
./waitpid_path
mokshitha_putti@MokshithaPutti:~/ossp-mini-shell$ abcxzy
abcxyz: command not found
mokshitha_putti@MokshithaPutti:~/ossp-mini-shell$ |
```

8. To Detect Variable References, Expand Values, Handle Undefined Variables, Update Tokens, Support Nested Variables, Test Expansion Logic.

To Identify Built-ins, Create Dispatch Table, Execute In-Process Commands, Handle Invalid Commands, Maintain State, Test Dispatch Logic.

```
mokshitha_putti@MokshithaPutti:~/ossp-mini-shell$ abcxyz
abcxyz: command not found
mokshitha_putti@MokshithaPutti:~/ossp-mini-shell$ cd ~/ossp-mini-shell
nano src/builtin.c
mokshitha_putti@MokshithaPutti:~/ossp-mini-shell$ gcc src/builtin.c -o builtin
./builtin
myshell> pwd
/home/mokshitha_putti/ossp-mini-shell
myshell> echo $USER
echo mokshitha_putti
myshell> help
Builtins: pwd help exit
myshell> |
```

```
mokshitha_putti@MokshithaPutti:~/ossp-mini-shell$ abcxyz
abcxyz: command not found
mokshitha_putti@MokshithaPutti:~/ossp-mini-shell$ cd ~/ossp-mini-shell
nano src/builtin.c
mokshitha_putti@MokshithaPutti:~/ossp-mini-shell$ gcc src/builtin.c -o builtin
./builtin
myshell> pwd
/home/mokshitha_putti/ossp-mini-shell
myshell> echo $USER
echo mokshitha_putti
myshell> help
Builtins: pwd help exit
myshell> xyz
xyz
myshell> echo $USER
echo mokshitha_putti
myshell> echo $ABC
echo
myshell> echo $HOME/$USER
echo /home/mokshitha_putti/mokshitha_putti
myshell> pwd
pwd
myshell> help
Builtins: pwd help exit
myshell> cd ..
cd ..
myshell> pwd
pwd
myshell> xyz
xyz
myshell> |
```

9. To Change Directories, Validate Paths, Update Working Directory, Handle Errors, Maintain Previous Directory, Test Navigation.

To Identify Built-ins, Create Dispatch Table, Execute In-Process Commands, Handle Invalid Commands, Maintain State, Test Dispatch Logic.

```

mokshitha_putti@MokshithaPutti:~/ossp-mini-shell$ cd ~/ossp-mini-shell
nano src/q9_shell.c
mokshitha_putti@MokshithaPutti:~/ossp-mini-shell$ gcc src/q9_shell.c -o q9
mokshitha_putti@MokshithaPutti:~/ossp-mini-shell$ ./q9
myshell> pwd
/home/mokshitha_putti/ossp-mini-shell
myshell> cd ..
myshell>
myshell> pwd
/home/mokshitha_putti
myshell> cd -
myshell> pwd
/home/mokshitha_putti
myshell> help
Built-in Commands:
cd pwd help exit
myshell> cd abc
cd: No such file or directory
myshell> xyz
Invalid Command: xyz
myshell>

```

10. To Retrieve Current Directory, Display Path, Process Exit Requests, Cleanup Resources, Save State, Verify Correct Behavior.

To Parse Export Syntax, Update Environment Variables, Validate Names, Handle Existing Variables, Export to Child Processes, Test Functionality.

```

mokshitha_putti@MokshithaPutti:~$ pwd
/home/mokshitha_putti
mokshitha_putti@MokshithaPutti:~$ echo $PATH
/home/mokshitha_putti/bin:/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/usr/lib/wsl/lib:/mnt/c/Program Files/WindowsApps/MicrosoftCorporationII.WindowsSubsystemForLinux_2.6.1.0_x64__8wekyb3d8bbwe:/mnt/c/Program Files/Common Files/Oracle/Java/javapath:/mnt/c/Program Files (x86)/Common Files/Oracle/Java/java8path:/mnt/c/Program Files (x86)/Common Files/Oracle/Java/javapath:/mnt/c/Windows/system32:/mnt/c/Windows:/mnt/c/Windows/System32/Wbem:/mnt/c/Windows/System32/WindowsPowerShell/v1.0:/mnt/c/Windows/System32/OpenSSH:/mnt/c/Program Files/Docker/Docker/resources/bin:/mnt/c/Program Files/Git/cmd:/mnt/c/Program Files/nodejs:/mnt/c/Program Files/MySQL/MySQL Shell 8.0/bin:/mnt/c/Program Files/Common Files/Oracle/Java/javapath:/mnt/c/Program Files (x86)/Common Files/Oracle/Java/java8path:/mnt/c/Program Files (x86)/Common Files/Oracle/Java/javapath:/mnt/c/Windows/system32:/mnt/c/Windows:/mnt/c/Windows/System32/Wbem:/mnt/c/Windows/System32/WindowsPowerShell/v1.0:/mnt/c/Windows/System32/OpenSSH:/mnt/c/Program Files/Docker/Docker/resources/bin:/mnt/c/Users/Mokshitha Putti/AppData/Local/Microsoft/WindowsApps:/mnt/c/Users/Mokshitha Putti/AppData/Local/Python/pythoncore-3.14-64/Scripts:/mnt/c/Users/Mokshitha Putti/AppData/Roaming/Python/Python314/Scripts:/mnt/c/users/mokshitha_putti/.local/bin:/mnt/c/users/mokshitha_putti/appdata/local/packages/pythonsoftwarefoundation.python.3.12_qbz5n2kfra8p0/localcache/local-packages/python312/scripts:/mnt/c/Users/Mokshitha Putti/AppData/Local/Programs/Microsoft VS Code/bin:/mnt/c/Users/Mokshitha Putti/AppData/Roaming/npm:/snap/bin
mokshitha_putti@MokshithaPutti:~$ exit

```

11. To Store Commands, Maintain History Buffer, Handle Capacity Limits, Retrieve Entries, Display History, Validate Consistency.

To Create Pipeline Structures, Store Commands, Maintain Execution Order, Connect Processes, Validate Pipeline Layout, Test Configurations.

12. To Create Pipes, Redirect Streams, Execute Connected Commands, Close Descriptors, Synchronize Processes, Test Data Flow.

To Support Multiple Pipes, Manage Descriptor Arrays, Launch Multiple Processes, Coordinate Execution, Cleanup Resources, Test Long Pipelines.

13. To Parse Input Redirection, Open Files, Redirect stdin, Handle Missing Files, Restore Streams, Test File Inputs.

To Parse Output Redirection, Create Output Files, Redirect stdout, Handle Permissions, Verify Output Data, Test Redirection.

14. To Parse Append Operator, Open Files in Append Mode, Redirect Streams, Preserve Existing Data, Validate Appends, Test Scenarios.

To Parse stderr Redirection, Redirect Error Streams, Capture Error Output, Verify File Contents, Handle Failures, Test Error Cases.

15. To Parse Combined Redirection, Duplicate File Descriptors, Merge Streams, Verify Ordering, Handle Edge Cases, Test Integrated Outputs.

To Parse Complex Commands, Combine Pipes and Redirections, Resolve Execution Order, Validate Syntax, Generate Execution Plans, Test Scenarios.

16. To Detect Background Jobs, Launch Non-Blocking Processes, Return Prompt Immediately, Store Job Information, Monitor Execution, Test Background Tasks.

To Create Job Table, Store Process Groups, Track States, Update Status, Remove Completed Jobs, Validate Job Records.

17. To Implement how to List Active Jobs, Display Status Information, Retrieve Job IDs, Format Output, Update State Information, Test Job Listings.

To design how to Identify Target Job, Transfer Terminal Control, Resume Processes,

Wait for Completion, Update Job State, Test Foreground Switching.

18. To Analyze Signal Delivery, Understand Signal Masks, Review Process Groups, Explore Terminal Signals, Test Signal Behavior.

To implement how to resume Stopped Jobs, Send Continue Signals, Update Job Status, Maintain Job Table, Verify Execution, Test Background Recovery.

19. To Register Handlers, Configure Signal Actions, Handle Interrupts, Manage Signal Context, Verify Stability, Test Signal Events.

To Capture SIGINT, Forward Signals, Protect Shell Process, Terminate Foreground Jobs, Update Job States, Test Interrupt Handling.

20. To Capture SIGTSTP, Suspend Processes, Update Job Table, Preserve State, Resume Later, Test Suspension Workflow.

To Create Process Groups, Assign Group IDs, Transfer Terminal Control, Restore Ownership, Coordinate Signals, Validate Behavior.

21. To Connect Modules, Validate Interfaces, Test End-to-End Execution, Handle Errors, Improve Reliability, Refactor Components.

To Detect Runtime Errors, Generate Messages, Log Failures, Handle Invalid Syntax, Recover Gracefully, Test Error Scenarios.

22. To Run Valgrind, Identify Leaks, Fix Memory Errors, Release Resources, Verify Cleanup, Document Findings.

To Design Test Cases, Create Test Scripts, Automate Execution, Verify Outputs, Compare Results, Generate Reports.

23. To Execute Large Pipelines, Launch Multiple Jobs, Measure Stability, Monitor Resources, Detect Failures, Analyze Performance.

To Document Architecture, Create Flow Diagrams, Explain Execution Paths, Describe Syscalls, Record Design Decisions, Review Documentation.

24. To Organize Source Code, Improve Build Scripts, Add README, Verify Repository Structure, Tag Releases, Validate Deliverables.

To Create Demo Scenarios, Validate Features, Rehearse Presentation, Demonstrate Pipelines, Demonstrate Signals, Final Project Review.